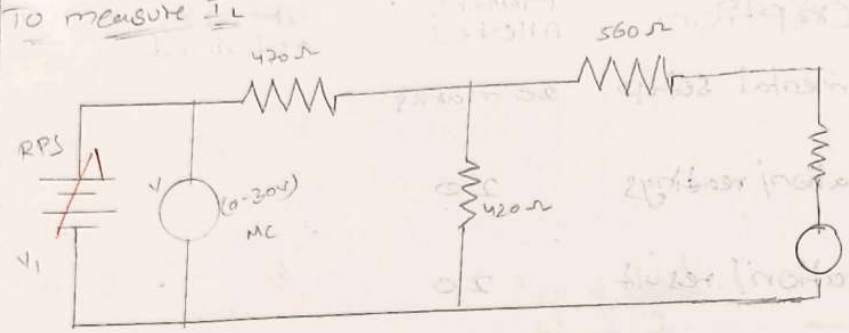
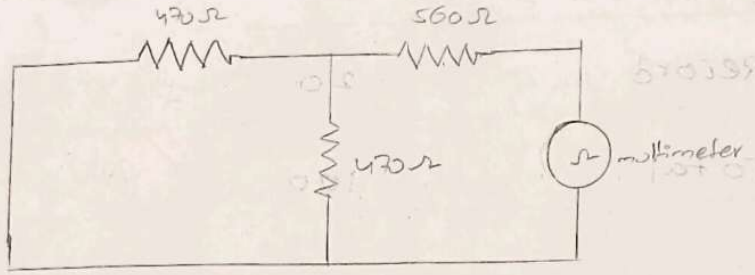
Diagram:

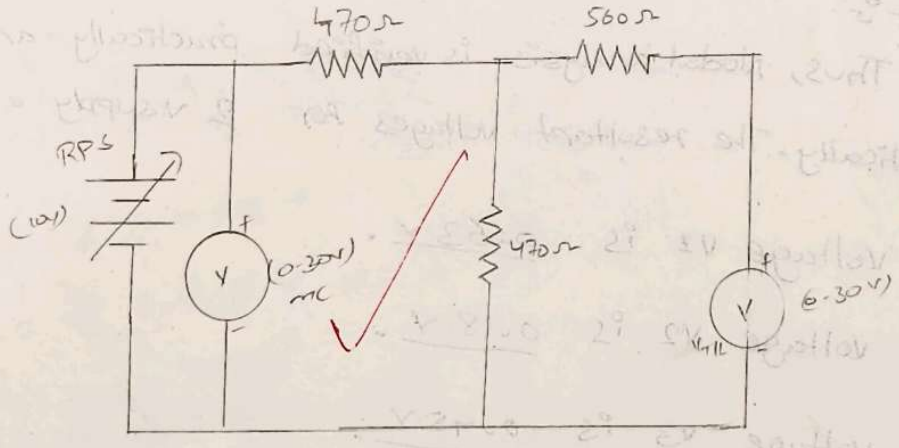
To measure I_L



To measure R_{TH} or R_N:



To measure V_{TH} or V_{OC}:



Ex.No :- 04

Date :- 07/02/26

Verification of Thevenin's Theorem

Aim :-

To verify Thevenin's theorem practically and theoretically for the given DC circuit.

Apparatus Required :-

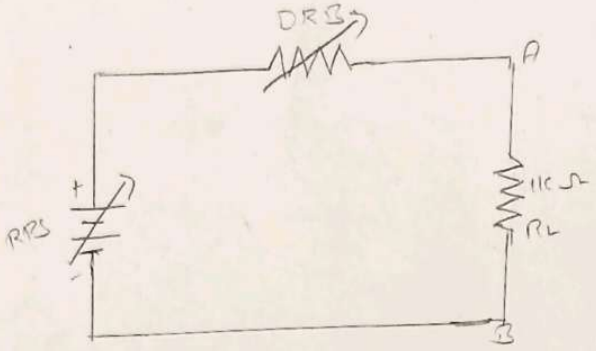
SL.No	APPARATUS	specification	quantity
1.	RPS	(0-30)V	1
2.	Voltmeter	(0-30V)MC	1
3.	Ammeter	(0-10mA)MC	1
4.	Resistors	470 Ω	2
		560 Ω	1
		1K Ω	1
5.	Bread board	–	1
6.	Multimeter	–	1

THEORY :-

Thevenin's Theorem :-

Thenvenin's theorem states that a linear two-terminal circuit can be replaced by an equivalent circuit consisting of a voltage source, $V_{Th} = V_{oc}$ in series with a resistor R_{Th}, where

Thevenin's equivalent circuit :-



Tabular column 1 :-

To measure I_L :-

Vi (volts)	I_L (AMPS)
10V	2.9A

Tabular column 2 :-

To measure R_{TH} or R_N :-

Vi (volts)	R_{TH} (ohm)
10V	769 Ω

Tabular column 3 :-

To measure V_{th} (or) V_{oc} :-

Vi (volts)	V_{th} (volts)
10	5.6

V_{th} is open-circuit voltage at the load terminals and R_{TH} is the input or equivalent resistance at the load terminals (AB) when the independent source are turned off".

Thenverin's voltage V_{TH} :-

It is the open-circuit voltage across the load terminals when all the sources are replaced by their interval resistance.

PROCEDURE :-

1. Make the connections as per circuit diagram 1.
2. Vary the RPS and set an input voltage of 10V.
3. Note down the voltmeter reading (V_1) and ammeter reading (I_L) in tabular coloumn 1.
4. Switch off the supply and make connections for circuit diagram 2.
5. Measure the theverin's resistance R_{TH} = Norton's resistance R_N.
6. Switch off the supply and make connections for circuit diagram 3.
7. Set an input voltage of 10V. in the RPS and note down the volt meter reading V_1 and ammeter reading $I_N = I_{SC}$ in tabular coloumn 3.

9. Set an input voltage of 10V in the RPS and note down the voltage reading V_1 and ammeter reading $I_N = I_{sc}$ in tabular column 4.

10. Draw thevenin's equivalent circuit and Norton's equivalent circuit as shown in circuit diagram 5 & 6.

11. Calculate the I_L value using the formula thevenin's theorem

$$I_L = V_{Th}/(R_{Th}+R_L)$$

Norton's theorem

$$I_L = I_N * R_N/(R_N+R_L)$$

12. Theoretically verify the Norton's theorem

Marks split up :-

S.No	Description	marks Alloted	marks obtained
1.	Experimental setup	20	20
2.	observation/Readings	20	20
3.	calculations/Results	20	18
4.	viva	20	18
5.	Record	20	18
6.	Total	100	94

RESULT :-

Thus, Thevenin's theorem is verified practically and theoretically

Q.S.V